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E2ETrADS: end-to-end transformer based autonomous driving system for adverse weather conditions

Sotirios Spanogianopoulos^{1*} and Kenan Ahiska²

Abstract

Adverse weather conditions, such as snow, heavy rain, fog, or limited illumination, pose significant challenges for autonomous vehicles (AVs) by degrading sensor reliability. This paper introduces E2ETrADS, an end-to-end transformer-based autonomous driving framework designed to operate robustly under such conditions. A comprehensive dataset was generated using the CARLA simulator, encompassing both nominal and adverse weather scenarios. The model is trained via imitation learning from an expert driver equipped with weather-adaptive MPC planner and PID controllers, enabling robust control under perception uncertainty. Experimental results demonstrate that E2ETrADS outperforms the TransFuser baseline in adverse conditions, exhibiting fewer infractions and improved lane adherence. The system dynamically adjusts vehicle speed to maintain control stability and adapts its control policies in response to sensor degradation, resulting in fewer missed turns and reduced lane invasions. Furthermore, E2ETrADS generalizes effectively to safety-critical long-tail scenarios, demonstrating human-like reasoning capabilities.

Keywords Autonomous driving, Transformer, CARLA, Adverse weather, Imitation learning

1 Introduction

Recent advances in deep learning have accelerated the development of fully autonomous vehicles (AVs), particularly in the areas of perception, decision making, and control. Although normal weather conditions have been extensively studied in the literature, maintaining reliable sensor performance and robust navigation under adverse weather conditions remain a major challenge [1, 2]. Environmental factors such as rain, fog, snow, and low illumination degrades sensor functionality: LiDAR systems

often produce sparse or noisy point clouds, reducing detection range and intensity [1, 2], while camera images may become blurred or occluded, impairing visual perception [3].

Sensor fusion is a solution to tackle these challenges; it integrates multimodal data such as cameras, LiDAR, radar, and GPS. Convolutional neural networks (CNNs), late fusion techniques, and rule-based systems are conventional sensor fusion methods; however, in adverse weather conditions, their performance is limited. Transformer-based architectures, on the other hand, are promising candidates for more robust solutions. Transformers benefit from a unified attention-based mechanism that can weigh different sensor data in space and time to capture inter-modal relationships and long-range dependencies for interpreting multiple contextual cues, which differs from CNNs that rely on localized receptive fields

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and rigid structures [4, 5]. It has been shown that TransFuser [6], which processes camera and LiDAR inputs simultaneously for trajectory prediction, outperforms CNN-based solutions, particularly in complex driving scenarios. BEVFormer employs bird's-eye-view spatial reasoning, and DETR adopts an advanced object detection approach to capture semantic information from multimodal inputs [7]. These examples position transformer-based architectures as promising navigation solutions for adverse weather conditions.

Collecting actual driving data is not always safe, cost-effective, or feasible; therefore, to study AVs, simulation tools play an important role. CARLA (Car Learning to Act) is one such platform for evaluating algorithms developed for perception, decision-making, and planning under various conditions, including dynamic traffic scenarios, varying illumination conditions, and adverse weather conditions. Furthermore, CARLA provides high-fidelity sensor models for semantic segmentation and RGB cameras, LiDAR, radar, IMU, and GNSS [8].

This paper proposes a transformer-based autonomous driving solution for adverse weather conditions and realistic traffic scenarios using the CARLA simulator. An imitation learning approach is used to train the transformer-based sensor fusion model, in which the actions of an expert driver along predetermined trajectories and under adverse weather conditions are replicated. As quantitative metrics, route completion, collision frequency, and infraction count are measured, and it is observed that, compared with the TransFuser baseline, the proposed model outperforms in fog, rain, and low-visibility scenarios, incurring significantly fewer infractions by internalizing behavioral strategies that generalize more effectively across varying weather conditions.

The structure of the paper is as follows. In Sect. 2, a literature review on autonomous driving (AD), challenges in AD due to adverse weather conditions, and imitation learning for autonomous driving is provided. In Sect. 3, the simulation environment setup and dataset, the proposed model architecture, and details of training using imitation learning are presented. In Sect. 4, the evaluation metrics and the performance of the proposed system under normal and adverse weather conditions in comparison with an expert autopilot and the TransFuser scheme are demonstrated. The paper concludes with Sect. 5.

2 Literature review

Conventionally, a modular pipeline (perception, localization, planning, and control) is constructed in autonomous driving systems, where each of these components operates independently but in coordination with one another through predefined interfaces [9]. For perception, sensor data from cameras, LiDAR, and radar, together with navigation sensors such as GPS and IMU,

are utilized, while SLAM-based methods are employed for localization. For planning MPC, and for control PID based solutions are widely used among various rule-based and/or optimization-based techniques. The advantage of the pipeline architecture lies in its modularity in terms of design, interpretation, and debugging; however, it also has drawbacks, such as limited adaptability to corner cases, including adverse weather conditions [10].

End-to-end deep learning approaches, where raw sensor input is mapped directly to control commands, have been proposed as an alternative to the conventional pipeline solution for autonomous driving. Bojarski et al. demonstrated the potential of end-to-end learning approaches for autonomous steering [11]. Codevilla et al. employed Conditional Imitation Learning for end-to-end autonomous driving [12]. Although the underlying philosophy of these studies is easy to comprehend, they lack transparency, require large annotated datasets, and cannot be readily generalized to long-tail scenarios, such as adverse weather conditions.

Autonomous navigation (perception) solutions fuse multimodal data to overcome the limitations associated with individual sensor weaknesses. Sensors on a single vehicle suffer from occlusions; cameras may experience poor contrast, LiDAR may produce sparse data, and radar lacks resolution in certain conditions [13, 14]. Bayesian inference is used to fuse state estimates or tracked objects from different sensors [15]; however, the assumptions of Gaussian noise characteristics and linearity limit these classical fusion schemes [16].

Deep learning-based navigation solutions can perform multi-sensor data fusion directly from raw sensor input without requiring the explicit design of a fusion algorithm. They are classified into three categories depending on the network input: early fusion (raw sensor data), intermediate fusion (feature maps), and late fusion (estimates or tracked objects) [17]. Among these, intermediate fusion is particularly attractive since it provides a balance between data-sharing overhead and resolution and preserves modality-specific characteristics during the feature alignment phase. Gao et al. proposed a CNN-based fusion approach for visual and depth data in object detection [18]. Chen et al. and Vora et al. proposed architectures that combine image data and point clouds for object detection [19, 20]. These methods, however, use local receptive fields with static sizes, which results in limited performance in capturing long-range dependencies under variable sensor reliability.

As a solution to these limitations, transformer-based autonomous navigation approaches have emerged. Transformers use self-attention mechanisms, making them well-suited for multimodal sensor data with consideration of long-range dependencies. Transformers were introduced for sequence modeling in natural language

processing (NLP) [4] and were subsequently applied to computer vision in methods such as DETR [21], Swin Transformer [22], and Vision Transformer [23], achieving superior performance compared to CNNs in object detection and segmentation. Transformers do not rely on spatial closeness and reweigh input data based on their contextual relevance using self-attention mechanisms; therefore, they are suitable for asynchronous, partially missing, or noisy sensor data [24]. Some studies employ transformers in early fusion (raw sensor data), while others utilize hierarchical attention mechanisms for deeper fusion of spatial and semantic cues [6].

Although their computational cost is significantly higher compared to conventional CNNs and the data required for training are not always easy to collect, transformer-based navigation methods appear suitable for autonomous driving due to their adaptability to sensor noise, occlusions, and missing data. Transfuser is a transformer-based autonomous navigation system fusing camera and LiDAR features [25], using a separate convolutional encoder to process camera images and a point-based encoder for LiDAR data prior to fusion. Transfuser captures more abstract spatial relationships across modalities, which allows it to outperform CNNs in terms of generalization and accuracy. DETR3D [26] and BEVFormer [7] combine sensor data in a unified bird's-eye-view (BEV) space, where self-attention mechanisms perform long-range spatial reasoning on fragmented data from multiple cameras and sparse LiDAR point clouds. SceneTransformer and GeTrans [27] work on spatial and temporal changes, which makes these methods more effective on sequential data, an advantageous property for trajectory behavior forecasting problems.

Adverse weather conditions, such as precipitation, fog, or limited illumination, significantly affect AV sensors. Cameras suffer from glare, motion blur, lack of adequate contrast, and occlusion [28]. LiDARs experience scattered pulses and noisy point clouds, resulting in up to a 50% reduction in effective range [2]. Although radars are more robust in adverse weather, they cannot provide

sufficient spatial resolution and suffer from interference in urban environments [29]. To tackle these issues, multi-modal solutions are widely preferred [30].

In adverse weather conditions, data collection is limited, costly, and potentially dangerous, which is why simulation tools such as CARLA (Car Learning to Act) have been widely used in AV research [8]. CARLA provides weather presets, such as WetCloudyNoon and HardRainSunset, in which the sensor models are updated, and offers a rich set of driving settings (10 towns and numerous roads) for studying generalizable techniques as can be seen in Fig. 1.

The simulation-to-reality gap is an important challenge. Approaches such as adversarial generation or feature transfer across weather conditions have been suggested as potential solutions [31]. Real-world adverse weather datasets are also available, such as Oxford RobotCar [32] and nuScenes [33]; however, they are limited in diversity and lack ground-truth synchronization.

In the control of an AV, the actions of human drivers or optimal controllers (throttle, braking, and steering commands) can be replicated using imitation learning (IL), and a learning-based model that maps sensor observations to control actions can be obtained [34]. This model is expected to act in a human-like manner, which can be safer than reinforcement learning-based solutions when tackling dense traffic or sharp turns. Many top-performing models have used IL with behavior cloning (BC) and dataset aggregation (DAGger) in the CARLA challenges [35].

3 Method

3.1 Model architecture

In this paper, the capabilities of the CARLA simulator to simulate adverse weather and illumination effects on AV sensors are utilized for training and testing an end-to-end transformer-based autonomous driving system (E2ET-rADS). The limitations of using a simulator are considered; however, diverse synthetic weather scenarios are constructed to build a generalizable solution which can



Fig. 1 CARLA scenes (a) sunny day (dry road) (b) rainy day (wet and reflective road)

be resilient when deployed. A multi-sensor AV configuration, consisting of an RGB camera and LiDAR, is considered. Weather-adaptive Model Predictive Control (MPC) planner and Proportional-Integral-Derivative (PID) controllers are employed to generate expert driving behavior, and, using imitation learning (IL), the E2ETrADS system is trained. The transformer-based navigation module at the center encodes each modality separately and generates low-level control actions. In Fig. 2, the methodology is demonstrated.

The front-facing RGB camera input is processed using a ResNet-34 network, from which 2D features are extracted. The LiDAR point cloud is encoded using a CNN-based BEV backbone and processed to capture spatial occupancy and scene geometry. The two modality-specific features are treated as separate tokens. Navigation commands, such as ‘go straight’ and ‘turn left/right,’ are treated as an additional token. These are encoded into a sequence of unified vector tokens (one token per modality and the navigation commands) before being fed to the transformer. A six-layer Transformer encoder is utilized in the E2ETrADS; at each layer, there are eight attention heads. A global self-attention mechanism is applied across all input tokens, allowing the model to capture cross-modality correlations and adjust attention dynamically based on contextual cues. At the output layer, a fully connected regression head generates the control commands—steering angle, throttle, and brake—which completes the end-to-end architecture of E2ETrADS.

Behavioral cloning (BC) is used to train the model. The expert driver is imitated, and at each instant, the sensor inputs (front-facing RGB camera and LiDAR) and

navigation commands are recorded for training. The control commands of the expert driver are generated by a hybrid weather-adaptive MPC trajectory planner and PID controller.

The weather-adaptive MPC controller employs a linear system model, where the vehicle states consist of the 2D position and yaw angle, to generate a long-term trajectory. The planner operates at 10Hz. The objective is to minimize the deviation from the reference trajectory while reducing control effort. The prediction horizon is set to 20, and the control horizon is set to 10. The weights in the cost function are adjusted according to weather and illumination conditions to optimize maneuverability and ensure operation within safe speed regimes.

Two PID controllers are employed in the low-level control design: one for longitudinal control and one for lateral control. From the trajectory generated by the weather adaptive MPC, waypoints with temporal annotations are generated. The longitudinal PID controller regulates the vehicle velocity to match the desired velocity. Based on the resulting velocity error, throttle and brake commands are computed. The desired velocity is determined using the current vehicle position, the next waypoint, and the time required to reach that waypoint. The parameters of the longitudinal PID controller are set to $K_p = 5.0$, $K_i = 0.5$, and $K_d = 1.0$. The lateral PID controller regulates the steering angle to track the desired heading. The parameters of the lateral PID controller are $K_p = 1.25$, $K_i = 0.75$, and $K_d = 0.3$. The control loop operates with a period of 10Hz. In summary, the PID control module computes the low-level control commands: steering angle, throttle, and brake inputs.

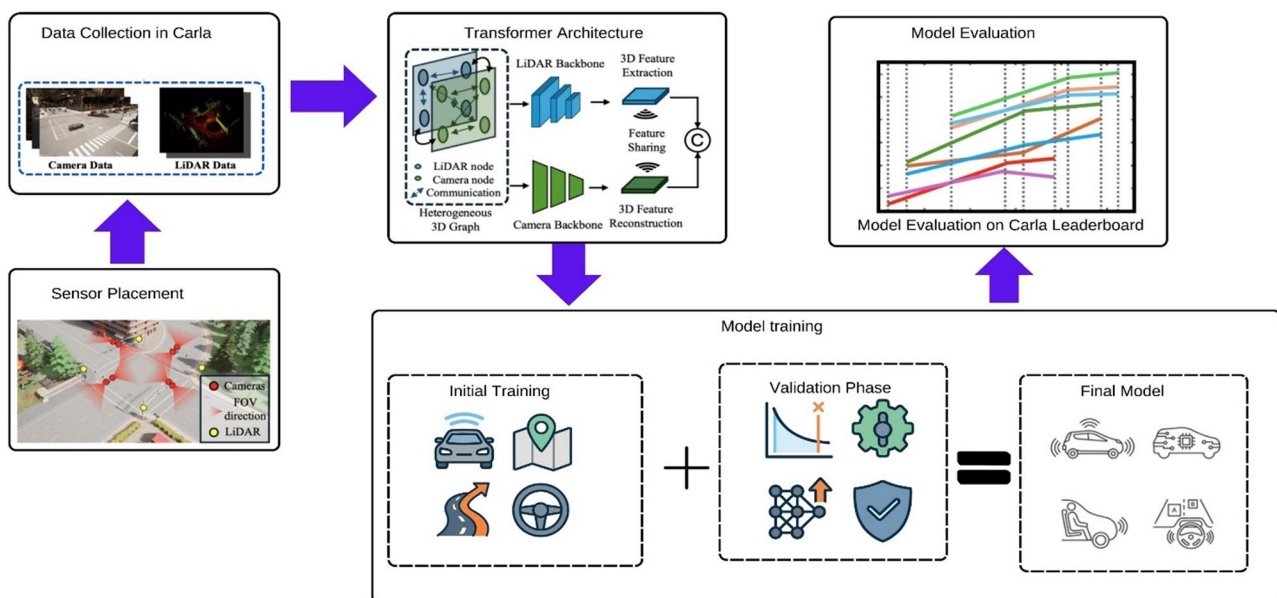


Fig. 2 E2ETrADS: data collection, transformer-based navigation and control

3.2 Simulation setup, dataset collection and training

The simulations are performed using CARLA 0.9.10 running in a Python 3.7 environment on Ubuntu 18.04 LTS. The system is equipped with an Intel Core i7-9700KF processor (8 cores, 3.6GHz), 32 GB of DDR4 RAM, and an NVIDIA GeForce RTX 2070 Super GPU with 8 GB of GDDR6 memory, along with a 500 GB SSD for system storage. The open-source TransFuser architecture (2022) is adopted and integrated seamlessly with the CARLA simulator. The required libraries are provided within a Conda environment, and necessary adjustments are made for CUDA 11.1 and PyTorch 1.9.0.

Two towns, Town01 and Town03, are used in the simulations; see Fig. 3. Town01, with its simpler layout, is selected for training as it provides stable data collection and efficient annotation, while Town03 is used for testing and evaluation due to its complex urban geometries, including intersections and elevation changes, as well as denser traffic, which are necessary to assess the robustness of the proposed framework.

To enhance the generalizability and robustness of E2ETrADS against behavioral complexity, simulations are augmented with dynamic vehicle and pedestrian behaviors using CARLA's Traffic Manager, including vehicles abruptly changing lanes and randomly braking, pedestrians suddenly crossing or being occluded, and vehicles running red lights. CARLA's weather presets—WetCloudyNoon, HardRainSunset, MidRainyNight, and WetSunset—are used to improve robustness against adverse weather conditions. These weather presets change the amount of illumination, fog, rain, and surface reflection, which affect the performance of the RGB camera and LiDAR.

The ego vehicle is equipped with an RGB camera (800×600 pixels, 90° FOV, 20Hz), a LiDAR (64 beams, 360° scans, 10Hz), and GPS and IMU. The sensor suite is shown in Fig. 4. Over approximately 200 hours of driving, 72,000 camera frames, 36,000 LiDAR scans, and 36,000

IMU measurements are recorded and associated with the vehicle pose, navigation commands, and control actions.

The dataset is partitioned into a 70/15/15 split for training, validation, and testing, respectively. The mean squared error (MSE) between predicted control inputs and expert control commands is minimized during training. The Adam optimizer is used for training, with the learning rate initialized at 1×10^{-4} . The batch size is 8, and the maximum number of epochs is set to 41, with early stopping applied if no improvement in the validation loss is observed. Regularization techniques, including dropout (0.3 across Transformer layers), label smoothing, and input data augmentation such as occlusions, are applied. Training is performed on an NVIDIA RTX 2070 Super GPU, with the total training time being 280 hours (approximately 7 hours per epoch).

4 Results

In the tests, the performance of E2ETrADS is compared against the TransFuser baseline and the expert model, which has knowledge of the vehicle states and the map used during imitation learning training. Four performance metrics are considered:

- **Route Completion:** The percentage of the vehicle completing the trajectory. The end conditions include exceeding the time limit, a collision or infraction, or irretrievable deviation from the route (higher is better).
- **Infraction Penalty:** Includes collisions with pedestrians, objects, or vehicles, as well as red-light violations and lane invasions. The vehicle starts with a score of 1, and for each violation, the score is multiplied by a factor less than 1, depending on the assigned penalty (higher is better).
- **Driving Score:** A metric calculated as the product of the route completion metric and the infraction penalty (maximum value 100, higher is better).



(a)



(b)

Fig. 3 CARLA towns (a) Town1 (b) Town3

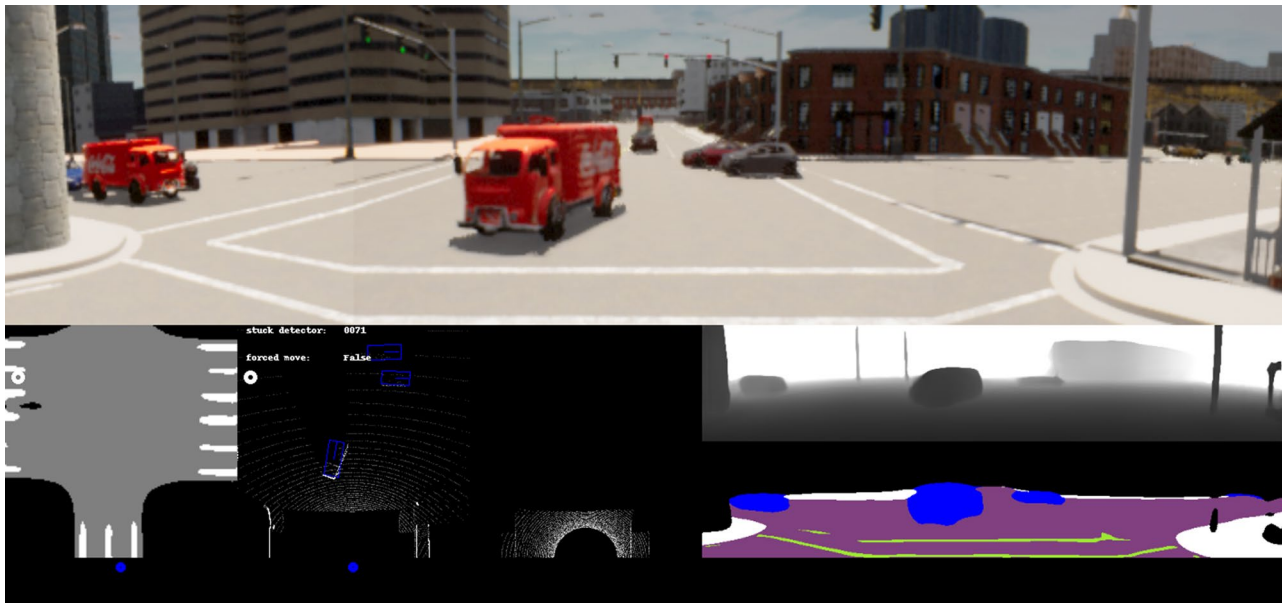


Fig. 4 Sensor suite of the ego vehicle

Table 1 Performance metrics under normal weather scenarios

Metric	Expert model	E2ETrADS	TransFuser
Route Completion (%)	88.5	82.2	85.3
Infraction Penalty	0.95	0.83	0.77
Driving Score	78.4	72.6	71.3
Control Smoothness	High	High	High

Table 2 Performance metrics under adverse weather scenarios

Metric	Expert model	E2ETrADS	TransFuser
Route Completion (%)	82.7	72.3	86.5
Infraction Penalty	0.894	0.88	0.71
Driving Score	74.5	66.3	61.2
Control Smoothness	High	High	Medium

- Control Smoothness: A qualitative metric assessed based on acceleration profiles over the entire trajectory (higher is better).

In Table 1, the performance metrics under normal weather scenarios are presented. E2ETrADS demonstrates fewer infractions, better control reliability, and improved route adherence, despite slightly lower route completion compared to the TransFuser baseline. The relatively lower route completion metric is attributed to the safety-prioritizing weather-adaptive MPC used in the imitation learning for the training of E2ETrADS. This results in lower speeds in E2ETrADS but enables smoother deceleration during turns and improved lane centering. Compared to the expert model, the TransFuser baseline incurs more infractions in complex intersections and dense urban areas.

In Table 2, the performance metrics under adverse weather scenarios are presented. The results demonstrate

the superiority of E2ETrADS in adverse weather conditions. Its speed is dynamically reduced to preserve control fidelity. Fewer infractions and better road adherence are observed compared to TransFuser, as the latter does not adjust its policies in response to unreliable sensor data. E2ETrADS missed fewer turns and committed fewer lane invasions, thereby preventing over-steering. The expert model and E2ETrADS have lower route completion scores, although their infractions are fewer compared to TransFuser. This indicates that both models reached the time limit specified in the route completion criteria. The lower route completion score of E2ETrADS is attributed to the reduced speed profiles employed by the model, which resulted in exceeding the time limit.

Trajectory plots demonstrate that E2ETrADS follows smoother trajectories, especially in roundabouts, turns, and intersections. E2ETrADS adjusts speed profiles based on visibility and road slipperiness, applying preemptive braking or wider turning radii under adverse weather conditions. The safety-prioritizing nature of E2ETrADS avoids aggressive route completion at the cost of potential infractions involving pedestrians, objects, and vehicles. E2ETrADS favors slower speed profiles to reduce risk in adverse weather conditions and preserve mission integrity.

5 Conclusions

In this paper, an end-to-end transformer-based autonomous driving system, E2ETrADS, is proposed. The system uses a transformer-based self-attention mechanism for multi-modal sensor input and generates control commands in an end-to-end fashion. Training is performed using the behavior cloning method, in which an expert

driver is imitated. The expert driver controller consists of weather-adaptive MPC planner and PID controllers and prioritizes safety over speed and aggressive control. Tests are conducted on the CARLA simulator under both normal and adverse weather conditions. The results demonstrate the superiority of E2ETrADS over the TransFuser baseline.

E2ETrADS does not rely solely on data-driven policies but also accounts for sensor resilience and weather conditions. It makes weather-aware adjustments to control actions, enabling a more deliberate drive even under reduced sensor and control performance. E2ETrADS operates at reduced speeds, resulting in a lower route completion score; however, infractions and rule violations are also reduced, even under adverse weather conditions. E2ETrADS is shown to be generalizable in safety-critical long-tail scenarios through its human-like reasoning acquired via behavioral cloning in these edge cases.

Despite their unified architecture, transformer-based end-to-end autonomous driving models face key barriers to deployment. The simulation-to-real gap limits generalization due to discrepancies between simulated and real environments [36, 37], while domain adaptation must address distributional shifts under multimodal and safety-critical constraints [38]. Moreover, sensor mismatch in noise, calibration, and synchronization can degrade robustness [39]. Future work will emphasize real-vehicle validation and incremental refinement of E2ETrADS to bridge these gaps.

Author contributions

S.S. made substantial contributions to the conception of the work, K.A. revised the work critically for important intellectual content, S.S. and K.A. approved the version to be published. All authors reviewed the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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